

Midterm review

We have a DC motor that has a no-load speed, ω_{nl} , of **1200 rpm**. At **800 rpm**, it is producing **0.6 Nm** of torque.

- What is the expected **stall torque** τ_s for this motor (in Nm)?
What is the **peak power** (in Watts)?
- Draw and properly label a plot of the **speed–torque** and **power–torque** curves for this motor, and show your calculations (draw the general trends using the key points that define the curves).

$$\tau = \tau_s \left(1 - \frac{\omega}{\omega_{nl}} \right)$$

Stall torque

$$\tau_s = \frac{\tau}{1 - \omega/\omega_{nl}} = \frac{0.6}{1 - \frac{800}{1200}} = \frac{0.6}{\frac{1}{3}} = 1.8 \text{ Nm}$$

Expected stall torque: 1.8 Nm

Peak power

Motor power $P = \tau \omega$.

For a linear DC motor, power is maximized at **half** speed and **half** stall torque:

$$\omega = \frac{\omega_{nl}}{2} = 600 \text{ rpm}, \quad \tau = \frac{\tau_s}{2} = 0.9 \text{ Nm}$$

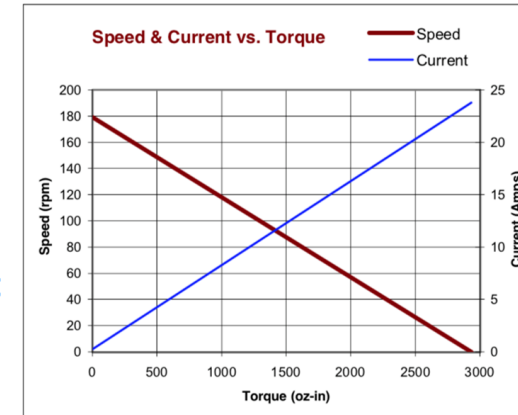
Convert 600 rpm to rad/s:

$$600 \text{ rpm} = \frac{600}{60} 2\pi = 10 \cdot 2\pi = 20\pi \approx 62.832 \text{ rad/s}$$

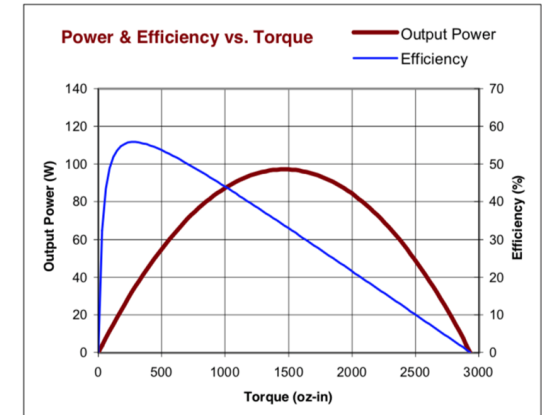
$$P_{\max} = \tau \cdot \omega = 0.9 \times 62.832 \approx 56.55 \text{ W}$$

(Equivalently, $P_{\max} = \tau_s \omega_{nl} / 4 = 1.8 \times (1200 \text{ rpm} \rightarrow 125.664 \text{ rad/s}) / 4 \approx 56.55 \text{ W}$.)

Peak power: 56.6 W (approx.)



All values are nominal. Specifications subject to change without notice. Graphs are shown for reference only.

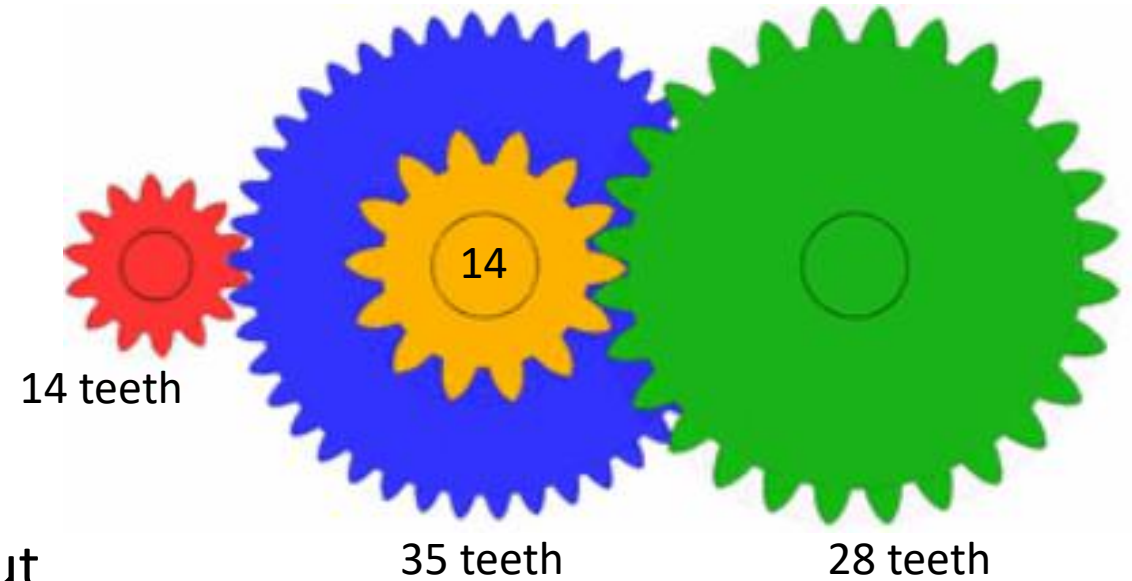


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If the red gear is driven with a speed of 200 rad/s and a torque of 2200 oz-in:

What is the output speed?

What is the output torque?



How fast would a car go if the green output gear is connected to a wheel with a diameter of 10 cm?

$$i_1 = \frac{\omega_{\text{red}}}{\omega_{\text{blue}}} = \frac{N_{\text{blue}}}{N_{\text{red}}} = \frac{35}{14} = 2.5$$

$$i_2 = \frac{\omega_{\text{orange}}}{\omega_{\text{green}}} = \frac{N_{\text{green}}}{N_{\text{orange}}} = \frac{28}{14} = 2$$

$$i = i_1 \times i_2 = 2.5 \times 2 = 5$$

$$\omega_{\text{out}} = \frac{\omega_{\text{in}}}{i} = \frac{200}{5} = \boxed{40 \text{ rad/s}}$$

$$T_{\text{out}} = T_{\text{in}} \times i = 2200 \times 5 = \mathbf{11000 \text{ oz}\cdot\text{in}} \text{ } (\approx \mathbf{77.5 \text{ N}\cdot\text{m}})$$

$$v = \omega r = 40 \times 0.05 = \boxed{2.0 \text{ m/s}}$$

$$2.0 \text{ m/s} = \boxed{7.2 \text{ km/h}} \approx \boxed{4.47 \text{ mph}}$$

What is 0xE9 in binary? If this binary number represents an 8-bit two's complement signed integer, what is its value in decimal?

Solution

Step 1. Convert hex to binary

Hex	Binary
E	1110
9	1001

$$0xE9 = 1110\ 1001_2$$

Step 2. Interpret as signed (two's complement)

Compute its magnitude:

1 Invert bits:

$$11101001 \rightarrow 00010110$$

2 Add 1:

$$00010110 + 1 = 00010111_2$$

3 Convert to decimal:

$$00010111_2 = 16 + 4 + 2 + 1 = 23$$

So the number is **-23**.

Representation	Value
Binary	11101001₂
Signed decimal (two's complement)	-23₁₀

What do the letters in PID control stand for? What is the principal responsibility of each in terms of adjusting the control response?

Letter	Full Term	Primary Role in Control
P	Proportional	Reduce error
I	Integral	Eliminates steady-state error
D	Derivative	Prevent overshoot

Our robot has a motor attached to a gearbox.

The motor has an encoder with **256 counts/revolution**.

The gearbox provides a **12:1** reduction from the motor to the output shaft.

A wheel of **80 mm diameter** is mounted on the output shaft.

Question: What is the **spatial resolution** of the encoder in terms of the motion of the wheel? (Specify your answer in {count/cm}.)

Solution

1. Counts per output shaft revolution

$$256 \times 12 = 3072 \text{ counts/rev (of the wheel)}$$

2. Wheel circumference

$$C = \pi D = \pi \times 80 \text{ mm} = 251.33 \text{ mm} = 25.133 \text{ cm}$$

3. Spatial resolution

$$\text{Resolution} = \frac{3072 \text{ counts}}{25.133 \text{ cm}} = 122.2 \text{ counts/cm}$$

We have a robotic car, and we want it to go to a target **150 mm** away.

We use a proportional controller with a gain of **2.5 percent per mm** (control output is % of max PWM).

How far will the robot travel (in millimeters) before the power to the motor drops to 60 percent?

Assume no integral or derivative action (**K_i = K_d = 0**).

Solution

Proportional control output:

$$u(\%) = K_p \cdot e \quad (\text{with } e = \text{distance to target})$$

When the motor command reaches the threshold $u_{\text{th}} = 60\%$:

$$e_{\text{th}} = \frac{u_{\text{th}}}{K_p} = \frac{60}{2.5} = 24 \text{ mm}$$

Distance already traveled:

$$d_{\text{travel}} = d_0 - e_{\text{th}} = 150 - 24 = \boxed{126 \text{ mm}}$$

We have a **PD controller** with a setpoint target of **100 counts** and gains of

$$K_P = 5.0, \quad K_D = 40$$

Our motor has currently traveled **70 counts**, and its speed is **1.5 counts per second**.

If the control variable is a **PWM value**, what is the **output of our controller**?

Solution

The general PD control law is:

$$u = K_P(e) + K_D \left(\frac{de}{dt} \right)$$

where

$$e = \text{setpoint} - \text{position}$$

and

$$\frac{de}{dt} = -(\text{speed})$$

(because as the motor moves forward, the error decreases).

Step 1. Compute the position error:

$$e = 100 - 70 = 30 \text{ counts}$$

Step 2. Compute the derivative of error:

$$\frac{de}{dt} = -1.5 \text{ counts/s}$$

Step 3. Substitute into PD equation:

$$u = (5.0)(30) + (40)(-1.5)$$

$$u = 150 - 60 = \boxed{90}$$

Port registers

- Ports are registers inside the microcontroller connected to pins
 - Allow for external reading and writing of certain registers
- ATmega328p ports are associated with 3 registers
 - **Data Direction Register** (e.g. DDRB): dictates whether pin is INPUT (0) or OUTPUT (1)
 - **Port Input Pin** (e.g. PINB): stores the current value of the pin
 - **Data Register** (e.g. PORTB): internal storage for port bit values
 - (if pin is INPUT the Data Register controls the Pullup resistor)
 - (see ATmega328p datasheet pg 75-6)

Port registers

- Data (PORT) and data direction (DDR) registers can directly be written to
 - PIN is read-only
- `DDRB = 0b00100000` // pin 5 is output, rest are input
 - In Arduino, the register names are macros to edit the register directly
- `PORTB = 0b00100100` // set pin 5 to high, pin 2 to pull-up input
- Use bit-masking to avoid accidentally changing other pins

Use a bitmask to set pins 3 and 7 to output (on register DDRB) if you don't know what the values of the other pins are.

Other topics to review

- Encoders
- Digital logic
- Arduino microcontrollers
- Arduino programming

Take home practice

- To change the angular velocity range on the BNO055 9-axis accelerometer to from the default to 500 degrees per second, the register `GYR_Config_0` needs bits 2:0 set to 010. Write the code you would use to do this without changing any of the other bits of that register. You can assume that `GYR_Config_0` is already a variable.